

Chapter 2

Preconception Nutrition



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Learning Objective

At the end of this chapter students be able to:

- ✓ *Summarize the reproduction process, including the critical hormones and their roles.*
- ✓ *List the conditions that may contribute infertility or poor pregnancy outcome.*
- ✓ *List the common nutrients, including the recommended intake level , for preconception period.*
- ✓ *Describe the role of body weight and lifestyle habits in preconception health.*
- ✓ *Identify nutritional effects of contraceptives*

Preconception Overview

- ✓ Unhealthy life style prior to conception may result in infertility and poor outcomes during pregnancy and may result in persistent health issues that may impact the baby for his or her entire life.

- ✓ Preconception period is the time prior to or between pregnancies and it is important time for women of childbearing age to be in health condition.
- ✓ It is a critical and often overlooked , stage of reproductive cycle.

Question

Why Nutrition during Preconception is important?

- Today, many of women of childbearing age spend preconception period in less than optimal health, and therefore they start pregnancies as such.
- For example, about 31% pregnant women are obese, about 69% do not take folic acid and an estimated 3% take medications that are known to cause birth defects

- ✓ A mother's nutritional status prior to conception, (preconception 3 months before), and immediately afterwards, (periconception 2–3 months after), is critical.
- ✓ The fetus is most vulnerable to nutritional deficiencies in the first trimester of pregnancy, often before a woman realizes that she is pregnant.
- ✓ There is evidence that poor maternal nutrition has both immediate (e.g. low birth weight) and long-term consequences.

- If conception occurs during the time of poor maternal nutrition or health status, the growth and development of the fetus, and the future health of the offspring may all be compromised.
- In extreme cases, suboptimal health prior to pregnancy can lead to pregnancy related complications, such as gestational diabetes, high blood pressure and may result in infant death or premature birth with associated birth defects and disabilities.

Chapter Introduction

- Adequate **health** & **nutrition** status are needed for successful reproduction
- Inadequate **health** & **nutrition** status may disrupt reproductive capacity

Preconception Overview

Key terms:

- **Fertility**—Actual production of children
- **Infertility**—Absence of production of children

Preconception Overview: Infertility

- ✓ ~15% of couples are infertile
- ✓ 40% of couples diagnosed as “infertile” will conceive within 3 years without use of technology
- ✓ Healthy couples have a **23% to 30%** chance of conception **within a given menstrual cycle.**

Preconception Overview: Miscarriage

Miscarriage: Loss of conceptus in 1st 20 weeks of pregnancy

- Causes of miscarriages:
 - Defect in fetus
 - Maternal infection
 - Structural abnormalities of uterus
 - Endocrine or immunological disturbances

Preconception Overview: Subfertility

Subfertility: Reduced level of fertility characterized by unusually long time for conception

✓ ~18% of couples are subfertile

✓ Examples:

- Having multiple miscarriages
- Sperm abnormalities
- Infrequent ovulation

Female Reproductive system and Hormones.

- ❖ Ovaries are --produce hormones that regulate female secondary sex Characteristics as well as
 - ✓ House of the ova (eggs).
- ❖ Fallopian tubes are a pair of long
 - ✓ transport ova from the ovary to the uterus and ,
 - ✓ transport sperm cells from the uterus to the released ova.
- ❖ During puberty females develop monthly menstrual cycles, to prepare an ovum to be fertilized by sperm

- ✓ Four major hormones are involved in the female menstrual cycle namely:- Follicle stimulating hormone (FSH), Luteinizing Hormone (LH), estrogen, and progesterone.

❖ The menstrual cycle phase

- ✓ During follicular phase, the brain releases FSH and LH , which travel to the ovaries.
 - ✓ In the ovaries, they stimulate grow of follicle and Ovulation.
- ✓ Estrogen levels also rise in response to the increase in FSH and LH and once enough estrogen released, FSH is turned off, limiting the number of follicles that mature.
- ✓ One follicle becomes dominant and continues to mature and eventually the others die.

- ✓ **The luteal Phase** typically begins halfway through the 28-day cycle.
- ✓ Estrogen ---stimulates the release of **LH**, which causes the follicle to release its egg from the ovary.
- ✓ The egg is captured by the fallopian tubes in the process known as **ovulation**.
- ✓ If the egg is not fertilized , the egg passes through the uterus and the lining of the uterus breaks down and is released , a process is known as **menstruation**.

Male RS and Hormones

- Three major hormones are involved in the male RS. FSH, LH, and testosterone.
- FSH makes sperm. LH is responsible for making **testosterone**, which is also necessary for making sperm.
- Finally, Testosterone necessary for the development of male characteristics.

Reproductive Physiology: Ova and Women

- ✓ Women born with life-time supply of ~7 million immature ova
- ✓ ~400-500 ova will mature & be released during fertile years
- ✓ Chromosomes in ova may be damaged by oxidation, radioactive particle exposure, & aging
- ✓ **Women >35 years more likely to have disorders related to chromosomal defects than younger women**

Sources of Disruptions in Fertility

- ✓ Adverse nutritional exposure
- ✓ Contraceptive use
- ✓ Severe stress
- ✓ Infection
- ✓ Tubal damage or other structural damage
- ✓ Chromosomal damage

Nutrition-Related Disruptions in Fertility

- Nutrient intake from food and dietary supplements, body fat, affects fertility primarily by:
 - Altering the environment on which eggs and sperm develop and
 - Modifying levels of hormones
- dietary habits, weight status, and supplement use can directly affect a woman's chances of conceiving.

Nutrition-Related Disruptions in Fertility

- ✓ Undernutrition
- ✓ Weight loss
- ✓ Obesity
- ✓ High exercise levels
- ✓ Intake of specific foods & food components

Undernutrition and Fertility

Chronic undernutrition

- primary effect: birth of small infants with high likelihood of death in the first year of life

Acute undernutrition

- associated with a dramatic decline in fertility that recovers when food intake does

Body Fat and Fertility

- ✓ Decreased fertility seen with low or high body fat due to alterations in hormones
- ✓ Estrogen & leptin
 - levels increased with high body fat & reduced with low body fat
 - both extremes lower fertility
- ✓ Infertility lower with BMI <20 or >30

Weight Loss and Fertility in Females

- ✓ Weight loss $>10-15\%$ of usual weight decreases estrogen
 - ✓ Results in amenorrhea, anovulatory cycles, & short or absent luteal phases
- ✓ Treatment with fertility drug Clomid not effective in underweight women

Weight Loss and Fertility in Males

- ✓ Studies from World War II showed 50% decrease in male fertility during **starvation**.
- ✓ Sperm **viability** & **motility** decrease with wt. 10 to 15% <normal & cease at wt >25% < normal

Exercise and Fertility

Table 2.4 Incidence of irregular or absent menstrual cycles in female athletes and sedentary women

	Incidence of Irregular or Absent Menstrual Cycles
Joggers (5 to 30 miles per week)	23%
Runners (over 30 miles per week)	34%
Long-distance runners (over 70 miles per week)	43%
Competitive bodybuilders	86%
Noncompetitive bodybuilders	30%
Volleyball players	48%
Ballet dancers	44%
Sedentary women	13%

Carotenemia and Fertility

- ✓ High intake of beta-carotene linked to amenorrhea & menstrual dysfunction
- ✓ Skin color & fertility return to normal within 2-6 weeks after discontinuation of beta-carotene

Preconception Iron Status, Fertility, & Pregnancy Outcome

- Rate of infertility lower in women who use iron supplements or iron from plant foods.
- Pre-pregnancy iron deficiency linked to preterm delivery & low iron status of infant

Caffeine and Fertility

- ✓ Caffeine appears to prolong time to conception
- ✓ Daily caffeine intake & reduction in conception is:
 - 300 mg results in ~27% ↓
 - 500 mg results in ~50% ↓

Table 2.5 Caffeine content of foods and beverages

Foods and Beverages	Caffeine (mg)
Coffee, 1 c	
Drip	137–153
Percolated	97–125
Instant	61–70
Decaffeinated	0.5–4.0
Tea, 1 c	
Brewed 5 minutes	32–176
Instant	40–80
Soft Drinks, 12 oz	
Mountain Dew	54
Coca-Cola	46
Diet Coca-Cola	46
Dr. Pepper	40
Pepsi-Cola	38
Diet Pepsi-Cola	37
Ginger ale	0
7-Up	0
Chocolate Products	
Cocoa, chocolate milk, 1 c	10–17
Milk chocolate, 1 oz	1–15
Chocolate syrup, 2 tb	4

Alcohol and Fertility

- ❖ Alcohol may decrease **estrogen & testosterone** levels or disrupt menstrual cycles
- ❖ Studies on weekly drinks consumed show:
 - 1-5 drinks → 39% ↓ in conception
 - >10 drinks → 66% ↓ in conception
- ✓ It is recommended that women restrict their alcohol intake while attempting pregnancy.

Other Factors Contributing to Infertility in Males

- **Antioxidant nutrients**
 - Protect sperm from oxidative damage
- **Vitamin D**—Low status related to infertility
- **Alcohol intake**—toxic effect on testes
- **Heavy metal exposure**
 - Lead—impacts testes & sperm
 - Mercury—decreases sperm & semen
- **Halogens**

Nutrition- related Side Effects of Contraceptives

Table 2.6 Nutrition-related side effects
of contraceptives

Oral Contraceptives

- Increased blood levels of HDL cholesterol (the “good” cholesterol)
- Increased blood levels of triglycerides and LDL cholesterol
- Increased risk of venous thromboembolism (blood clots)
- Decreased blood levels of vitamin B₁₂
- Increased blood levels of copper

Contraceptive Injections (Depo-Provera)

- Weight gain
- Increased blood levels of LDL cholesterol and insulin
- Decreased blood levels of HDL cholesterol
- Decreased bone density

Contraceptive Implants (Norplant)

- Weight gain

Cont...

❖ In preparation for healthy pregnancy a women can establish the following habits:

- ✓ Achieve and maintain healthy body weight
- ✓ Choose an adequate and balanced diet
- ✓ Be physically active
- ✓ Receive regular medical care
- ✓ Manage chronic condition
- ✓ Avoid harmful influences

Nutritional Exposures Before and Very Early in Pregnancy that Disrupt Fetal Growth and Development

Folate Status Prior to Conception and Neural Tube Defects

- ✓ Inadequate folate very early in pregnancy can cause *neural tube defects* (NTDs).
- ✓ These defects develop within 21 days after conception or before many women even know they are pregnant
- ✓ Encouraging women to consume folic acid, a highly absorbable, synthetic form of this B vitamin.

Weight Status

- Being underweight increases the risk of
 - maternal complications during pregnancy and
 - delivery of small and early newborns
- Obesity increases the risk of clinical complications during pregnancy and delivery of newborns with **neural tube defects** or **excessive body fat**

Nutrient Status

- ✓ Insufficient **folate** intake increases the risk of neural tube defects
- ✓ Excessive **vitamin A** intake increases the risk of developing facial and heart abnormalities
- ✓ High blood levels of **lead** increase the risk of mental retardation
- ✓ **Iodine deficiency** early in pregnancy increases the risk of impaired mental and physical development
- ✓ **Iron deficiency** increases the risk of early delivery and development of iron deficiency

Cont...

Alcohol

- Regular intake of alcohol increases the risk of *fetal alcohol syndrome* and *fetal alcohol effects*

Diabetes

- Poorly controlled blood glucose levels early in pregnancy increase the risk of
 - fetal malformations,
 - excessive infant size at birth, and
 - development of diabetes in the offspring

Recommended Dietary Intakes for Preconceptional Women

It is recommended that women who may become pregnant

1. Consume 400 mcg of folate from fortified grain products, vegetables, fruits, or supplements
2. Take no more than 5000 IU of vitamin A
3. limit or omit alcohol-containing beverages
4. Iron 18-45mg/d
5. 140-150mcg of iodine
6. 1000mg calcium

THANK YOU